

Implementing Power-Law in LBM C++ Program

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1 Derivations & Procedure

Here we derive the implicit equation for the relaxation time τ at each lattice site k ,

$$\tau = \frac{1}{2} + \frac{3m}{\rho} \left(\frac{Q}{\tau} \right)^{n-1}$$

which is solved numerically at every site to recover the local, shear-rate-dependent value of τ required by a power-law non-Newtonian collision model. We then derive an analytic benchmark solution for a force-driven channel flow,

$$u_x(y) = U_{\max} \left[1 - \left(\frac{2|y - N_y/2|}{N_y} \right)^{(n+1)/n} \right]$$

which we use to validate the solver.

1.1 Derivation for Relaxation Time

The relaxation time in a standard Newtonian LBM simulation is fixed, since the kinematic viscosity ν is constant. However, for a power law fluid, the apparent viscosity μ depends on the local shear rate $\dot{\gamma}$, which itself depends on the (initially unknown) local value of τ . This creates an implicit relationship that must be solved at every site, every time-step. .

We start from equation (39) in the LBM review by V. Dzanic et al., which relates the strain rate tensor D_{ij} to the non-equilibrium part of the populations f_{α}^{neq} ,

$$D_{ij} = -\frac{1}{2\rho\tau c_s^2} \sum_{\alpha} \xi_{\alpha i} \xi_{\alpha j} f_{\alpha}^{\text{neq}}, \quad \dot{\gamma} = \sqrt{2\text{tr}(\mathbf{D}^2)} \quad (\text{L39})$$

This is convenient because f_{α}^{neq} is already computed as part of the collision step.

To simplify the bookkeeping, we define the tensor S_{ij} as the second moment of f_α^{neq} ,

$$S_{ij} = \sum_{\alpha} c_{\alpha i} c_{\alpha j} f_{\alpha}^{\text{neq}} \quad (1)$$

where we have also switched notation for the discrete velocity from ξ_α to $\mathbf{c}_\alpha$ to match our lattice conventions. Substituting (1) into (L39) and specialising to lattice units in D_2Q_9 (where $c_s^2 = 1/3$), the strain rate tensor reduces to

$$D_{ij} = -\frac{3}{2\rho\tau} S_{ij} \quad (2)$$

We now want $\dot{\gamma}$ in terms of S_{ij} . Since D_{ij} is symmetric (as S_{ij} is, by construction, symmetric), the trace of $\mathbf{D}^2$ is simply

$$\text{tr}(\mathbf{D}^2) = \sum_{i,j} D_{ij} D_{ij} = \frac{9}{4\rho^2\tau^2} \sum_{i,j} S_{ij} S_{ij} \quad (3)$$

Substituting this into the definition of $\dot{\gamma}$ from (L39) gives

$$\dot{\gamma} = \frac{3}{\rho\tau} \sqrt{\frac{\sum_{i,j} S_{ij} S_{ij}}{2}} = \frac{Q}{\tau} \quad (4)$$

where we have collected everything that does not depend on τ into a single site-local quantity,

$$Q := \frac{3}{\rho} \sqrt{\frac{1}{2} \sum_{i,j} S_{ij} S_{ij}},$$

Next we bring in the constitutive law for the fluid. Equation (32) from the LBM review by V. Dzanic et al. gives the power-law relation between apparent viscosity and shear rate,

$$\mu = m \dot{\gamma}^{n-1} \quad (\text{L32})$$

where m is the consistency index and n the flow behaviour index ($n < 1$ shear-thinning, $n > 1$ shear-thickening, $n = 1$ recovering the Newtonian case with $m = \mu$). Substituting (4) into (L32) gives the apparent viscosity purely in terms of the relaxation time,

$$\mu = m \left(\frac{Q}{\tau} \right)^{n-1} \quad (5)$$

Finally, we connect ν back to τ through the standard LBM viscosity relation. Equation (37) from the LBM review by V. Dzanic et al. states

$$\nu = c_s^2 \left(\tau - \frac{1}{2} \right) \frac{\Delta x^2}{\Delta t} \quad (\text{L37})$$

which, again in lattice units for D_2Q_9 , simplifies to

$$\nu = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \quad (6)$$

Since the kinematic and apparent viscosities are related by $\mu = \rho\nu$, we can equate (6) with (5) and rearrange for τ . This yields our implicit equation for the local relaxation time,

$$\tau = \frac{1}{2} + \frac{3m}{\rho} \left(\frac{Q}{\tau} \right)^{n-1} \quad (7)$$

1.2 Derivation for Analytic Solution

To validate the power-law collision model derived above, we compare against an analytic solution for a simple, well-characterised flow: steady, force-driven flow of a power-law fluid in a rigid channel.

We consider a 2D force-driven flow with body force $\mathbf{g} = (g_x, 0)$ acting in a channel of height N_y , with physical coordinates (x, y) . We impose four standard simplifying assumptions:

1. Steady flow ($\frac{\partial}{\partial t} = 0$).
2. Fully developed flow, i.e. no variation along the channel ($\frac{\partial}{\partial x} = 0$).
3. Unidirectional flow, i.e. no cross-stream velocity component ($\mathbf{u} = (u_x, 0)$).
4. Incompressible flow ($\rho = \text{const}$).

Continuity. Under these assumptions the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_x)}{\partial x} + \frac{\partial(\rho u_y)}{\partial y} = 0,$$

is satisfied identically: every term vanishes by assumptions 1–4.

Momentum balance. In the x -direction, the momentum equation is

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} \right) + \rho g_x$$

The left-hand side vanishes entirely: the first term by steadiness, and the remaining two because $u_y = 0$ and $\frac{\partial u_x}{\partial x} = 0$. Similarly τ_{xx} , which depends only on $\frac{\partial u_x}{\partial x}$, does not contribute. This leaves the balance between the pressure gradient, the shear stress gradient, and the body force,

$$\frac{\partial p}{\partial x} = \frac{\partial \tau_{yx}}{\partial y} + \rho g_x$$

In the y -direction, the same argument (with $u_y = 0$ throughout) gives

$$\begin{aligned} \rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) &= -\frac{\partial p}{\partial y} + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right) + \rho g_y \\ \Rightarrow \frac{\partial p}{\partial y} &= 0 \end{aligned}$$

which tells us that $p = p(x)$ only. Combining the two momentum balances, we are left with a single governing equation for the shear stress profile,

$$\frac{dp}{dx} = \frac{\partial \tau_{yx}}{\partial y} + \rho g_x \quad (8)$$

The left-hand side of (8) is a function of x alone, while τ_{yx} in general depends on y ; for the equation to hold at every (x, y) , both sides must in fact equal the same constant.

Pressure gradient. Considering the left-hand side first,

$$\frac{dp}{dx} = c_0 \implies p(x) = c_0 x + c_1 \quad (9)$$

for constants c_0, c_1 . Since we are studying a purely *force*-driven flow (i.e. the flow is sustained by the body force g_x alone, with no externally imposed pressure drop along the channel), we require $p(x) = \text{constant}$, and hence $c_0 = 0$.

Shear stress profile. With $c_0 = 0$, equation (8) becomes an ODE for the shear stress alone,

$$\frac{\partial \tau_{yx}}{\partial y} + \rho g_x = 0 \implies \tau_{yx}(y) = -\rho g_x y + c_2 \quad (10)$$

for some constant c_2 . The flow is symmetric about the channel centreline $y = N_y/2$, so by symmetry the shear stress must vanish exactly at the centreline. Imposing $\tau_{yx}(N_y/2) = 0$ fixes $c_2 = \rho g_x N_y/2$, giving the linear shear stress profile

$$\tau_{yx}(y) = \rho g_x \left(\frac{N_y}{2} - y \right). \quad (11)$$

Note that this result is entirely independent of the constitutive law of the fluid: it follows purely from the momentum balance and symmetry, and holds for Newtonian and power-law fluids alike. The rheology only enters once we relate τ_{yx} to the velocity gradient. At this step we could in future use a different constitutive law other than the power law.

Constitutive law and velocity profile. We now introduce the power-law constitutive relation,

$$\tau_{yx} = m \left| \frac{du_x}{dy} \right|^{n-1} \frac{du_x}{dy}, \quad (12)$$

From (11), τ_{yx} is positive for $y < N_y/2$ and negative for $y > N_y/2$; by (12), du_x/dy must have the same sign, meaning the velocity increases towards the centreline from both walls, as expected for a driven channel flow. We solve the problem on the lower half of the channel, $y < N_y/2$, where all quantities are positive, and use the symmetry of the profile to extend the result to the full domain $y \in [0, N_y]$ afterwards.

For $y < N_y/2$, both sides of (12) are positive, so the absolute value can be dropped and we can equate (11) with (12) directly,

$$m \left(\frac{du_x}{dy} \right)^n = \rho g_x \left(\frac{N_y}{2} - y \right) \implies \frac{du_x}{dy} = \left[\frac{\rho g_x}{m} \left(\frac{N_y}{2} - y \right) \right]^{1/n}. \quad (13)$$

Integrating with respect to y gives the velocity profile up to a constant,

$$u_x(y) = -\frac{n}{n+1} \left(\frac{\rho g_x}{m} \right)^{1/n} \left(\frac{N_y}{2} - y \right)^{(n+1)/n} + c_3. \quad (14)$$

We fix c_3 using the no-slip condition at the lower wall, $u_x(0) = 0$, which gives

$$c_3 = \frac{n}{n+1} \left(\frac{\rho g_x}{m} \right)^{1/n} \left(\frac{N_y}{2} \right)^{(n+1)/n},$$

and substituting back into (14) yields the velocity profile on the lower half of the channel,

$$u_x(y) = \frac{n}{n+1} \left(\frac{\rho g_x}{m} \right)^{1/n} \left[\left(\frac{N_y}{2} \right)^{(n+1)/n} - \left(\frac{N_y}{2} - y \right)^{(n+1)/n} \right]. \quad (15)$$

Full profile. By the symmetry of the problem about $y = N_y/2$, the solution on the upper half of the channel is obtained simply by replacing $(N_y/2 - y)$ with $(y - N_y/2)$. Combining both halves into a single expression valid across the whole channel $y \in [0, N_y]$, we arrive at the analytic velocity profile for force-driven channel flow of a power-law fluid,

$$u_x(y) = \frac{n}{n+1} \left(\frac{\rho g_x}{m} \right)^{1/n} \left[\left(\frac{N_y}{2} \right)^{(n+1)/n} - \left| \frac{N_y}{2} - y \right|^{(n+1)/n} \right] \quad (16)$$

We can introduce the maximum velocity $U_{\max}$ as the value of the velocity at the centreline $y = N_y/2$,

$$U_{\max} = \frac{n}{n+1} \left(\frac{\rho g_x}{m} \right)^{1/n} \left(\frac{N_y}{2} \right)^{(n+1)/n}$$

to write our expression in (16) as,

$$u_x(y) = U_{\max} \left[1 - \left(\frac{2|y - N_y/2|}{N_y} \right)^{(n+1)/n} \right] \quad (17)$$